

PA2 - DFAs

1. When defining an alphabet Σ , the symbols of Σ can be repeated.

☐ True

☐ False

2. A language L consists of all the possible combinations of symbols from an alphabet Σ .

☐ True

☐ False

3. A deterministic finite automata never has several transitions for the same input.

☐ True

☐ False

4. A Deterministic Finite Automata (DFA) can have infinite states.

☐ True

☐ False

5. The set of final (accept) states of a DFA can be empty.

☐ True

☐ False

6. The start state of a DFA can also be a final (accept) state.

☐ True

☐ False

7. All DFAs have an equivalent transition table.

☐ True

☐ False

8. The string 111111 belongs to the language $\{w \in \{1\}^* \mid |w| \text{ is odd}\}$

☐ True

☐ False

9. The string 110110110 belongs to the language $\{w \in \{0,1\}^* \mid n_0(w) \text{ is even}\}$

☐ True

☐ False

10. The string 101010 belongs to the language $\{x10y \mid x \text{ and } y \text{ are strings of 0's and 1's}\}$

☐ True

☐ False
